

Autonomous and Non-Autonomous AI Systems

Part 1: Semi-supervised learning with HITL (Human-in-the-Loop) Systems

by Steven Miers

◆ Component Guide				
Node	Op	In	Out	Role
◆ Product	Auto	Model plug	Predictions	ML Eng.
■ UI/API	Users*	Predictions	Results	SW Eng.
■ Data	Auto	Results	Feedback	Data Eng.
○ Curation	Human	Feedback	Curated	Curator
◆ Pipeline	Auto	Curated	Train data	Data Eng.
◆ Training	Auto	Train data	Artifact	ML Eng.
○ Release	Human	Artifact	Approved	Governance

* Users = data source only. ○ = Human Gate

► Agentic AI in HITL

● Data Collection

Agents query users, synthesize behavioral signals, generate synthetic labels (RLAIF).

⊙ NOT a Classifier

Agentic collection does NOT change HITL/HOTL classification. It changes what feeds INTO the Data Curation Service.

▶ Still Gated

An agentic collector in HITL still routes to a human curator. The agent is a tool — not the gate.

Node Classification

● LOOP MODE

Automated service; runs without human decision

⊙ HUMAN GATE

Human must approve before flow continues

● DATA / SURFACE

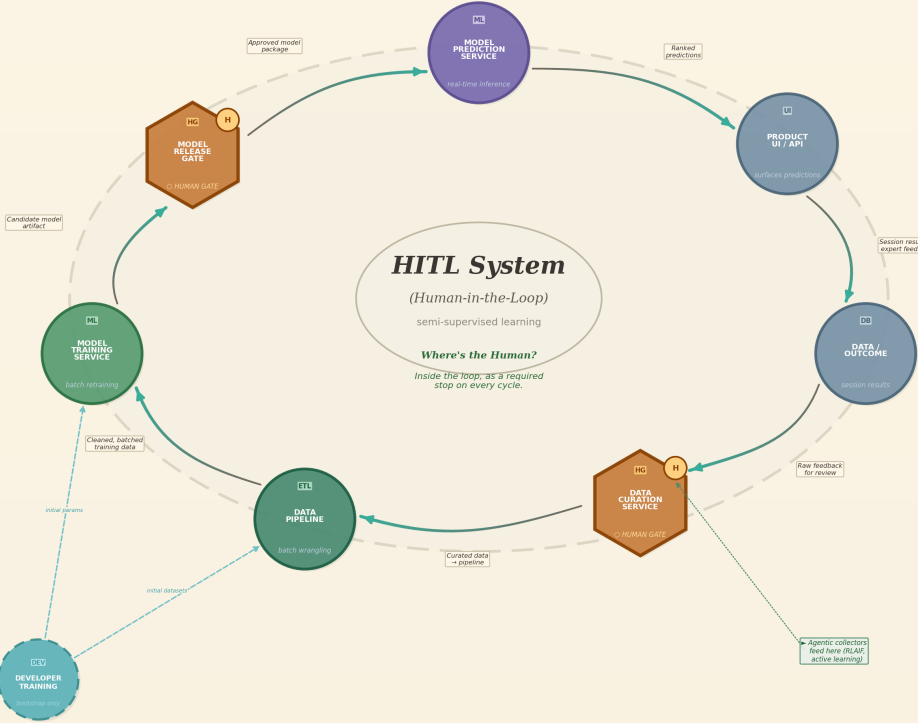
States or surfaces data; human contact incidental

● PIPELINE / TRAINING

Batch wrangling, model retraining, assembling

○ BOOTSTRAP

One-time setup; bypassed after first run



Where's the Human?

Not the End Users...

End users generate signal — they are a data source, not a gate.

The Gatekeepers

Decides what happens to data next:
► Curator: filters training data
► Reviewer: approves releases

```
$ the_math.sh  
  
users      = 1,000,000  
curation_team = 2  
  
# the 2-person team is  
# where the loop lives.
```

HITL Failure Modes

▲ Throughput Bottleneck

Humans cap pipeline rate.
94% of MLP practitioners cite annotation as binding constraint.

▲ Annotator Bias

Systematic disagreement by demographics & cultural context.
Mitigated by inter-annotator agent.

▲ Cost at Scale

RLHF comparisons expensive.
RLAIF + Constitutional AI are now mainstream alternatives.

◆ Autonomy Debt

1 Definition

"The gap between what your system could learn continuously and what it actually learns given human bandwidth."

2 As Data Scales

Hundreds + tens of thousands of matchable items.
The debt compounds.

3 Three Bottlenecks

► Curation = throughput cap
► Governance = deploy blocker
► Tuesday batches = stale models

4 Key Insight

The humans who make the system trustworthy **also make it slow.**

Many production AI systems are loop-shaped, but not closed-loop. Human feedback, curation, and governance introduce delay, filtering, and control boundaries that prevent true autonomy.